

rAllEvenDigits

The **recursive** function that returns either 1 or 0 according to whether or not all the digits of the positive integer argument number num are even. For example, if the argument num is 2468, then the function should return 1; and if the argument num is 2345, then 0 should be returned. Write the recursive function in two versions. The function rAllEvenDigits1() computes and returns the result. The function rAllEvenDigits2() computes and returns the result through the parameter result using call by reference. The function prototypes are given below:

```
int rAllEvenDigits1(int num);
void rAllEvenDigits2(int num, int *result);
```

A sample program template is given below to test the functions:

```
#include <stdio.h>
#define INIT_VALUE 999
int rAllEvenDigits1(int num);
void rAllEvenDigits2(int num, int *result);
int main()
{
    int number, result=INIT_VALUE;

    printf("Enter a number: \n");
    scanf("%d", &number);
    result = rAllEvenDigits1(number);
    if (result == 1)
        printf("rAllEvenDigits1(): yes\n");
    else if (result == 0)
        printf("rAllEvenDigits1(): no\n");
    else
        printf("rAllEvenDigits1(): error\n");
    result=INIT_VALUE;
    rAllEvenDigits2(number, &result);
    if (result == 1)
        printf("rAllEvenDigits2(): yes\n");
    else if (result == 0)
        printf("rAllEvenDigits2(): no\n");
    else
        printf("rAllEvenDigits2(): error\n");
    return 0;
}
int rAllEvenDigits1(int num)
{
    /* Write your code here */
}
void rAllEvenDigits2(int num, int *result)
{
    /* Write your code here */
}
```

Some sample input and output sessions are given below:

(1) Test Case 1:
Enter a number:
5
rAllEvenDigits1(): no
rAllEvenDigits2(): no

(2) Test Case 2:
Enter a number:

```
2468
rAllEvenDigits1(): yes
rAllEvenDigits2(): yes
```

(3) Test Case 3:

Enter a number:

```
2345
rAllEvenDigits1(): no
rAllEvenDigits2(): no
```

(4) Test Case 4:

Enter a number:

```
280
rAllEvenDigits1(): yes
rAllEvenDigits2(): yes
```